



Department of Electrical and Electronics Engineering Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: B.E, V Semester & EEE

Course Code & Title: EE3503 & Control Systems

Name of the Faculty member: Mr. A. Guna, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit I / Control system: Open loop and Closed loop
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO1
- **Activity Chosen:** One minute paper
- **Justification:**

The basic block diagram of control system has various components and the students understand the type of system through various examples. After teaching the concept, I thought of conducting this activity for making the students to give the difference between open and closed loop system which enhance the learning level of the students and as a teacher I can evaluate the understanding level of the students.

- **Time Allotted for the Activity:** (5 Minutes - Total time)
 - 2 Minute - Instruction about the activity
 - 2 Minute -Time given for the students to complete
 - 1 Minute - Collecting the paper
- **Details of the Implementation:**

The Purpose and Nature of the activity is explained to the students. Students were given 1 Minute of time to write in a piece of paper about the Concepts they have learnt in that day of class in few lines or points in bullet form. Papers are collected from the students and an assurance is given to the students that the concept will be discussed again in the next class. After teaching the concept, students were asked to take a sheet of paper and write the concepts they understood about open and closed loop system.

Total Strength is 64; Reporter: Myself

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about difference between open and closed loop system
- ✓ Then I told them to write as much as they can within a short period of 1 time.
- ✓ Finally, I collected the papers from the students

- **CO – PO / PSO mapping:**

CO	PO1	PO8	PO10	PO12	PSO3
CO1	3	1	1	1	2

- **PO / PSO mapped:**

Innovative practice	PO1	PO8	PO10	PO12	PSO3
	3	1	1	1	2
Justification for correlation	Students apply the mathematical techniques such as differentiation and Integration to mathematically model the given system	Students apply their moral and ethical principles while doing this activity	Students submit a clear, well-written statement in this activity	Student recognizes the need of analysis and able to clearly explain why it is important for the new development in the control system area	Students will be able to develop the mathematical model of the system which is required for the industrial automation of systems

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

The students enjoyed this One Minute Paper Activity. After the class many students requested me to insist other Course faculty members to conduct this activity. I have taken this feedback to by Head of the Department

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students actively participated in the One-minute paper activity. As a teacher I could know the real students feeling, their understanding level from this activity. This greatly helped me to correct my flaws in my teaching process and able to find what are all the concepts I have to retake in the class.

- ❖ ***Challenges faced in implementation:***

Still few students were hesitant to convey their points in paper. Few students have not taken this activity seriously and had given empty paper

References:

- ❖ **<https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-one-minute-paper>**

Signature of Faculty Member

HOD



Department of Electrical and Electronics Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE

Course Code & Title: EE3503 Control Systems

Name of the Faculty member (s): Mr. A. Guna, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit II /Root Locus Technique
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO9
- **Activity Chosen:** MATLAB Simulation

- **Justification:**

MATLAB provides you a huge number of toolboxes to plot graphs, visualize complex datasets. Students use MATLAB to perform complex scientific calculations, it also lets the students to visualize and implement complex mathematical functions. With the help of MATLAB simulation Students can determine the stability of the given system using Bode Plot in an easy way.

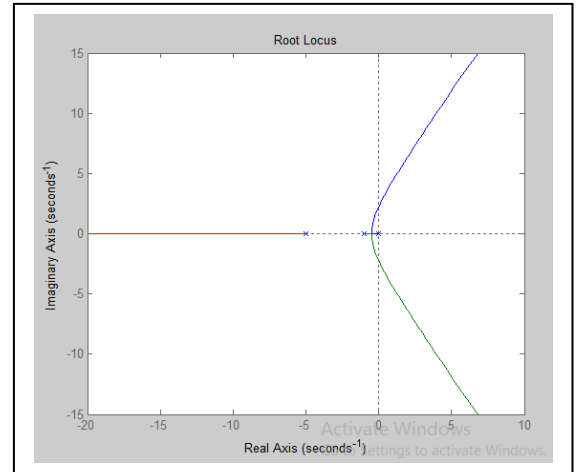
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**
 - ✓ All the students were taken to Control and Instrumentation Lab and the procedure to simulate the given system is explained to them
 - ✓ Students individually simulated the Root Locus of the given system
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO4	PO5	PSO3
CO2	3	3	2	2	2

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO4	PO5	PSO3
	3	3	2	2	2
Justification for correlation	Student applying their engineering knowledge to understand the concept of stability analysis of a given system	Students apply mathematical concepts to determine the stability of the given system	Students analyze the stability of the system and draws necessary conclusion	Students use MATLAB simulation software for determining time response of the system and their proficiency in MATLAB simulation is improved	Student analyze the performance of the given system using MATLAB software Package

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students find this simulation is the easier way to determine the stability of the given system.

Students also assured to continue the simulation using Online MATLAB Simulation

- ❖ **Benefit of the practice:**

The stability of a higher order system or any complicated system can be determined using MATLAB simulation

Student understood the Root Locus concept clearly and it is reflected during the discussion session.

- ❖ **Challenges faced in implementation:**

Few late bloomer students felt difficult in coding and simulation

References:

<https://in.mathworks.com/support/learn-with-matlab-tutorials.html>

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Course Code & Title: EE3503 & Control Systems

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Innovative Practice Description

- **Unit / Topic:** Unit IV / Controllability – Observability
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO21
- **Activity Chosen:** Flipped Classroom
- **Justification:**

This topic is chosen, since it can be easily understood by students. Students can improve their understanding on this topic by this activity. More time for discussion is given the class unlike traditional learning. This activity greatly helps the student to know the concept much deeper.

- **Time Allotted for the Activity:** 25 Minutes

- **Details of the Implementation:**

Each team was asked to do literature survey about the given topic. They should include at least 5 Reference material about the given topic for which the material is already shared. Students in a team were asked to present their topics. Queries will be asked by the students to the presenting team

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO10	PO12	PSO3
CO1	3	3	1	1	2

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO10	PO12	PSO3
	3	3	1	1	2
Justification for correlation	Students apply mathematical concepts to derive the state model for linear and time variant systems and to compute the solution of state and output equation	Students apply engineering mathematics and computations to solve state space model of the given system	Students will be able to effectively communicate their ideas in this activity	Student recognizes the need of analysis and able to clearly explain why it is important for the new development in the control system area.	Students will be able to analyse the controllability and observability of the system required for the industrial automation of systems

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

The students enjoyed this Flipped classroom Activity. After the class many students requested me to insist other Course faculty members to conduct this activity. I have taken this feedback to by Head of the Department

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students actively participated in the flipped classroom activity. As a teacher I could know the real students feeling, their understanding level from this activity. This greatly helped me to clear all the student doubt and make them more involved in the learning process

- ❖ ***Challenges faced in implementation:***

Still few students were hesitant to convey their points and queries. Few students have not taken this activity seriously and remain quite in the class

References:

- ❖ <https://www.panopto.com/blog/7-unique-flipped-classroom-models-right/>

Signature of Faculty Member

HOD